

Daogao Liu

GOOGLE RESEARCH

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PROFESSIONAL EXPERIENCE

Google Research, Mountain View, CA 2026.01 - Present
Research Scientist

- **Evaluation Lead (Gemini Thinking) & Post-Training Data/Eval:** Spearheaded RL and SFT data curation, synthesis, and evaluation methodologies for next-gen Gemini releases across **knowledge worker, web development, reasoning, cybersecurity, and SWE**.
- **Multi-Agent Systems & RSI:** Designed and scaled **multi-agent workflows** and **recursive self-improvement (RSI)** pipelines to push frontier autonomous reasoning and agentic capabilities.
- **Rapid Impact:** *Promoted within three months of joining*, following the delivery of agentic evaluation suites and training-data pipelines that drove measurable capability improvements across flagship model launches.

Google Research, Mountain View, CA 2024.12 - 2025.12
Visiting Postdoc Researcher (*Pivoted from pure theory to large-scale empirical LLM research*)

- **Urania — Privacy-Preserving Usage Analytics for LLMs** (*Project Lead*; [COLM 2025](#), [Google Research Blog](#)): Architected an end-to-end differentially private clustering and LLM-summarization framework to extract insights from chatbot conversations; productized by engineering teams and actively used internally at Google.
- **VaultGemma — Differentially Private 1B LLM** (*Core Contributor*; [Tech Report](#), [Google Research Blog](#)): Pretrained a 1B-parameter open language model with formal differential privacy guarantees, establishing a new Pareto frontier for private foundation models.
- **Gemini — IMO Gold Medal (2025)** (*Data Contributor*; [Official Blog](#)): Contributed reasoning training data to the Gemini Deep Think model that achieved gold-medal standard at the International Mathematical Olympiad, directly inspiring my transition to the Gemini post-training team.

Apple MLR, Cupertino, CA Summer, 2023/2024
Research Intern

- Developed near-optimal algorithms for private optimization, resolving several open questions in sample and computational complexity.
- Published 10 papers at top-tier machine learning venues (NeurIPS, ICML, ICLR, COLT), including an ICLR Spotlight improving upon the work at Google Brain.

Google Brain, Remote Spring 2023
Research Intern

- Focused on optimization and differential privacy with publications at top-tier conferences.
- Pioneered a novel private non-convex optimization framework, achieving NeurIPS Spotlight.

EDUCATION

University of Washington, Seattle, WA 2020 - 2024
Ph.D. in Computer Science Advised by [Yin Tat Lee](#)
Thesis: “Advancing Differentially Private Optimization: Efficiency, Utility, and Applications”

Tsinghua University, Beijing, China 2016 - 2020
B.S. in Physics and Mathematics Advised by [Jian Li](#)
Thesis: “Stochastic Optimization: Approximation Algorithms on Metric Spaces”

PUBLICATIONS

* denotes equal contribution; most theoretical works are in (reverse) alphabetical order.

- [1] Badih Ghazi, Ravi Kumar, **Daogao Liu**, and Pasin Manurangsi. Nearly Linear-Time User-Level DP-SCO with Optimal Rates. *Accepted to COLT 2026. [ArXiv link](#)*

- [2] Syamantak Kumar, **Daogao Liu**, Kevin Tian, and Chutong Yang. Private Geometric Median in Nearly-Linear Time. *Accepted to NeurIPS 2025. ArXiv link*
- [3] Andrew Lowy*, **Daogao Liu***. Differentially Private Bilevel Optimization: Efficient Algorithms with Near-Optimal Rates. *Accepted to NeurIPS 2025. ArXiv link*
- [4] Da Yu, Edith Cohen, Badih Ghazi, Yangsibo Huang, Pritish Kamath, Ravi Kumar, **Daogao Liu**, and Chiyuan Zhang. Scaling Embedding Layers in Language Models. *Accepted to NeurIPS 2025. ArXiv link*
- [5] Weijia Shi, Akshita Bhagia, Kevin Farhat, Niklas Muennighoff, Jacob Morrison, Evan Pete Walsh, Dustin Schwenk, Shayne Longpre, Jake Poznanski, Allyson Ettinger, **Daogao Liu**, Margaret Li, Mike Lewis, Wen-tau Yih, Dirk Groeneveld + 8 more authors. FlexOLMo: Open Language Models for Flexible Data Use. *Accepted to NeurIPS 2025. Draft link*
- [6] **Daogao Liu**, Edith Cohen, Badih Ghazi, Peter Kairouz, Pritish Kamath, Alexander Knop, Ravi Kumar, Pasin Manurangsi, Adam Sealfon, Da Yu, and Chiyuan Zhang. Urania: Differentially Private Insights into AI Use. *Accepted to COLM 2025. ArXiv link*
- [7] Guy Kornowski, **Daogao Liu**, and Kunal Talwar. Improved Sample Complexity for Private Nonsmooth Nonconvex Optimization. *Accepted to ICML 2025. ArXiv link*
- [8] **Daogao Liu** and Kunal Talwar. Adaptive Batch Size for Privately Finding Second-Order Stationary Points. *Accepted to ICLR 2025, Spotlight. ArXiv link*
- [9] Yangsibo Huang*, **Daogao Liu***, Lynn Chua, Badih Ghazi, Pritish Kamath, Ravi Kumar, Pasin Manurangsi, Milad Nasr, Amer Sinha, and Chiyuan Zhang. Unlearn and Burn: Adversarial Machine Unlearning Requests Destroy Model Accuracy. *Accepted to ICLR 2025. ArXiv link*
- [10] Weijia Shi, Jaechan Lee, Yangsibo Huang, Sadhika Malladi, Jieyu Zhao, Ari Holtzman, **Daogao Liu**, Luke Zettlemoyer, Noah A. Smith, and Chiyuan Zhang. Muse: Machine unlearning six-way evaluation for language models. *Accepted to ICLR 2025. ArXiv link*
- [11] Hilal Asi, **Daogao Liu**, and Kevin Tian. Private Stochastic Convex Optimization with Heavy Tails: Near-Optimality from Simple Reductions. *Accepted to NeurIPS 2024. ArXiv link*
- [12] Hilal Asi, Tomer Koren, **Daogao Liu**, and Kunal Talwar. Private Online Learning via Lazy Algorithms. *Accepted to NeurIPS 2024. ArXiv link*
- [13] Andrew Lowy, **Daogao Liu**, and Hilal Asi. Faster Algorithms for User-Level Private Stochastic Convex Optimization. *Accepted to NeurIPS 2024. ArXiv link*
- [14] Lynn Chua, Badih Ghazi, Yangsibo Huang, Pritish Kamath, **Daogao Liu**, Pasin Manurangsi, Amer Sinha, and Chiyuan Zhang. Mind the Privacy Unit! User-Level Differential Privacy for Language Model Fine-Tuning. *Accepted to COLM 2024. ArXiv link*
- [15] Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, **Daogao Liu**, Terra Blevin, Danqi Chen, and Luke Zettlemoyer. Detecting Pretraining Data from Large Language Models. *Accepted to ICLR 2024. ArXiv link*
- [16] Gavin Brown, Krishnamurthy Dvijotham, Georgina Evans, **Daogao Liu**, Adam Smith, and Abhradeep Thakurta, Private gradient descent for linear regression: Tighter error bounds and instance-specific uncertainty estimation. *Accepted to ICML 2024. ArXiv link*
- [17] Hilal Asi, **Daogao Liu**, User-level Differentially Private Stochastic Convex Optimization: Efficient Algorithms with Optimal Rates. *Accepted to AISTATS 2024. ArXiv link*
- [18] Arun Ganesh, **Daogao Liu**, Sewoong Oh, Abhradeep Thakurta. Private (Stochastic) Non-Convex Optimization Revisited: Second-Order Stationary Points and Excess Risks. *Accepted to NeurIPS 2023, Spotlight. ArXiv link*

- [19] Yair Carmon, Arun Jambulapati, Yujia Jin, Yin Tat Lee, **Daogao Liu**, Aaron Sidford, Kevin Tian. ReSQueueing Parallel and Private Stochastic Convex Optimization. *Accepted to FOCS 2023. ArXiv link*
- [20] Sivakanth Gopi, Yin Tat Lee, **Daogao Liu**, Ruoqi Shen, Kevin Tian. Algorithmic Aspects of the Log-Laplace Transform and a Non-Euclidean Proximal Sampler. *Accepted to COLT 2023. ArXiv link*
- [21] Hu Fu, Jiawei Li, **Daogao Liu**. Pandora Box Problem with Nonobligatory Inspection: Hardness and Improved Approximation Algorithms. *Accepted to STOC 2023. ArXiv link*
- [22] Sivakanth Gopi, Yin Tat Lee, **Daogao Liu**, Ruoqi Shen, Kevin Tian. Private Convex Optimization in General Norms. *Accepted to SODA 2023. ArXiv link*
- [23] Yaonan Jin, **Daogao Liu**, Zhao Song. Super-resolution in High Dimension: (Nearly) Linear Running Time and Sample Complexity. *Accepted to SODA 2023. ArXiv link*
- [24] Ziqi Wang, Yuexin Wu, Frederick Liu, **Daogao Liu**, Le Hou, Hongkun Yu, Jing Li, Heng Ji. Augmentation with Projection: Towards an Effective and Efficient Data Augmentation Paradigm for Distillation. *Accepted to ICLR 2023. ArXiv link*
- [25] Xuechen Li*, **Daogao Liu***, Tatsunori Hashimoto, Huseyin A Inan, Janardhan Kulkarni, Yin Tat Lee, Abhradeep Guha Thakurta. When Does Differentially Private Learning Not Suffer in High Dimensions? *Accepted to NeurIPS 2022. ArXiv link*
- [26] Sivakanth Gopi, Yin Tat Lee, **Daogao Liu**. Private convex optimization via exponential mechanism. *Accepted to COLT 2022. ArXiv link*
- [27] **Daogao Liu**. Better Private Algorithms for Correlation Clustering. *Accepted to COLT 2022. ArXiv link*
- [28] Jian Li, **Daogao Liu**. Multi-token Markov Game with Switching Costs. *Accepted to SODA 2022. ArXiv link*
- [29] Janardhan Kulkarni, Yin Tat Lee, **Daogao Liu**. Private Non-smooth ERM and SCO in Subquadratic Steps. *Accepted to NeurIPS 2021, Spotlight. ArXiv link*
- [30] Haotian Jiang, Jian Li, **Daogao Liu**, Sahil Singla. Algorithms and Adaptivity Gaps for Stochastic k-TSP. *Accepted to ITCS 2020. ArXiv link*
- [31] **Daogao Liu**. More efficient Algorithms for Stochastic Diameter and Some Unapproximated Problems in Metric Space. *Accepted to COCOON 2019.*

PROGRAM COMMITTEES SERVICE AND REVIEWING

- Area Chairs: ICLR 2025, ICML 2026, COLM 2026, TPDP 2023-2026
- Reviewers: SODA 2026/2025/2023, COLT 2025, ICML 2025, COLM 2025, AISTATS 2025, NeurIPS 2025/2024/2023, STOC 2024, ALT 2024, FOCS 2022/2021, RANDOM 2020
- Journals: Operations Research, Theoretical Computer Science, Information and Computation

SELECTED AWARDS

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| • Apple Scholars in AI/ML, Ph.D. Fellowship. | 2024 |
| • Paul G. Allen School First-Year Ph.D. Fellowship | 2020 |
| • Ye Qisun Nomination Award (The highest honor of Physics undergraduate), Tsinghua | 2020 |
| • Scholarship for Comprehensive Excellence, Tsinghua University | 2019 |